Quantification: 543_z13
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	6.3636	6	2.2923	count
ring_length_r30_px	headline	311	331	58.9016	px
ring_length_r60_px	reported	923.727	912	139.505	px
owned_length_px	ownership-dependent	2910.11	2471.87	1677.59	px
edge_count	ownership-dependent	127.454	107	81.1669	count
mean_edge_length_px	ownership-dependent	24.4052	22.6631	3.8624	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.03752	0.035343	0.0085221	px^-1
ring_density_r60	normalised	0.042509	0.0434941	0.00703491	px^-1
roots_per_100px_perimeter	normalised	3.113	3.2433	0.901	per 100 px
ring_attached_length_r30_px	network v2	245.546	260	60.0889	px
ring_attached_length_r60_px	network v2	658.818	740	201.961	px
ring_length_w_r30_px	network v2	352.991	376.534	65.1676	px
ring_length_w_r60_px	network v2	1047.51	1055.01	156.474	px
sholl_crossings_r10	network v2	6.8182	7	2.2724	count
sholl_crossings_r20	network v2	11.0909	12	2.5477	count
sholl_crossings_r30	network v2	13.4545	13	2.1616	count
ring_width_mean_r30_px	width	7.1679	7.3794	0.5203	px

Per field
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measure	value	units
analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	41377	px
canalicular_length_density_per_px	0.0402463	px^-1
junction_count	926	count
junction_density_per_px2	0.0009007	px^-2
lacuna_count	11	count
lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	231	count

mean_component_length_px	179.121	px
median_component_length_px	24	px
field_length_density_w_per_px	0.045718	px^-1
field_density_without_flagged_per_px	0.0402463	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	256	px
unattached_component_count	13	count
canalicular_length_density_attached_per_px	0.0402463	px^-1
field_length_density_with_unattached_per_px	0.0404953	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6274	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	41377	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	295	912	3742.82	172	21.7606	7.4007
2	7	354	951	4691.25	207	22.6631	6.6482
3	8	331	876	1524.6	62	24.5903	6.4844
4	3	238	858	1293.3	52	24.8712	7.7768
5	8	347	1047	5499.27	246	22.3548	7.8264
6	5	353	942	2173.93	107	20.3171	7.3794
7	8	337	1051	2942.97	131	22.4654	6.521
8	4	274	875	2471.87	98	25.2232	7.4497
9	5	232	750	1685.55	54	31.2139	7.0377
10	6	244	703	669.985	21	31.904	6.6364
11	11	416	1196	5315.69	252	21.094	7.6863

Normalised per lacuna

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lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	205.037	8499	22530	1	1	0.03471	0.0404794	2.4386
2	151.297	6982	17707	1	1	0.0507018	0.0537076	4.6267
3	205.037	8219	20045	1	1	0.0402725	0.0437017	3.9017
4	139.681	6734	18999	1	1	0.035343	0.0451603	2.1477
5	223.823	8900	23453	1	1	0.0389888	0.0446425	3.5742
6	141.338	6516	17817	1	1	0.0541743	0.0528708	3.5376
7	254.936	9810	24446	1	1	0.0343527	0.0429927	3.138
8	278.593	10387	26403	1	1	0.0263791	0.0331402	1.4358
9	195.723	8083	21794	1	1	0.0287022	0.0344131	2.5546
10	184.995	7835	21305	1	1	0.0311423	0.0329969	3.2433
11	301.806	10961	27498	1	1	0.0379527	0.0434941	3.6447

Network v2 per lacuna

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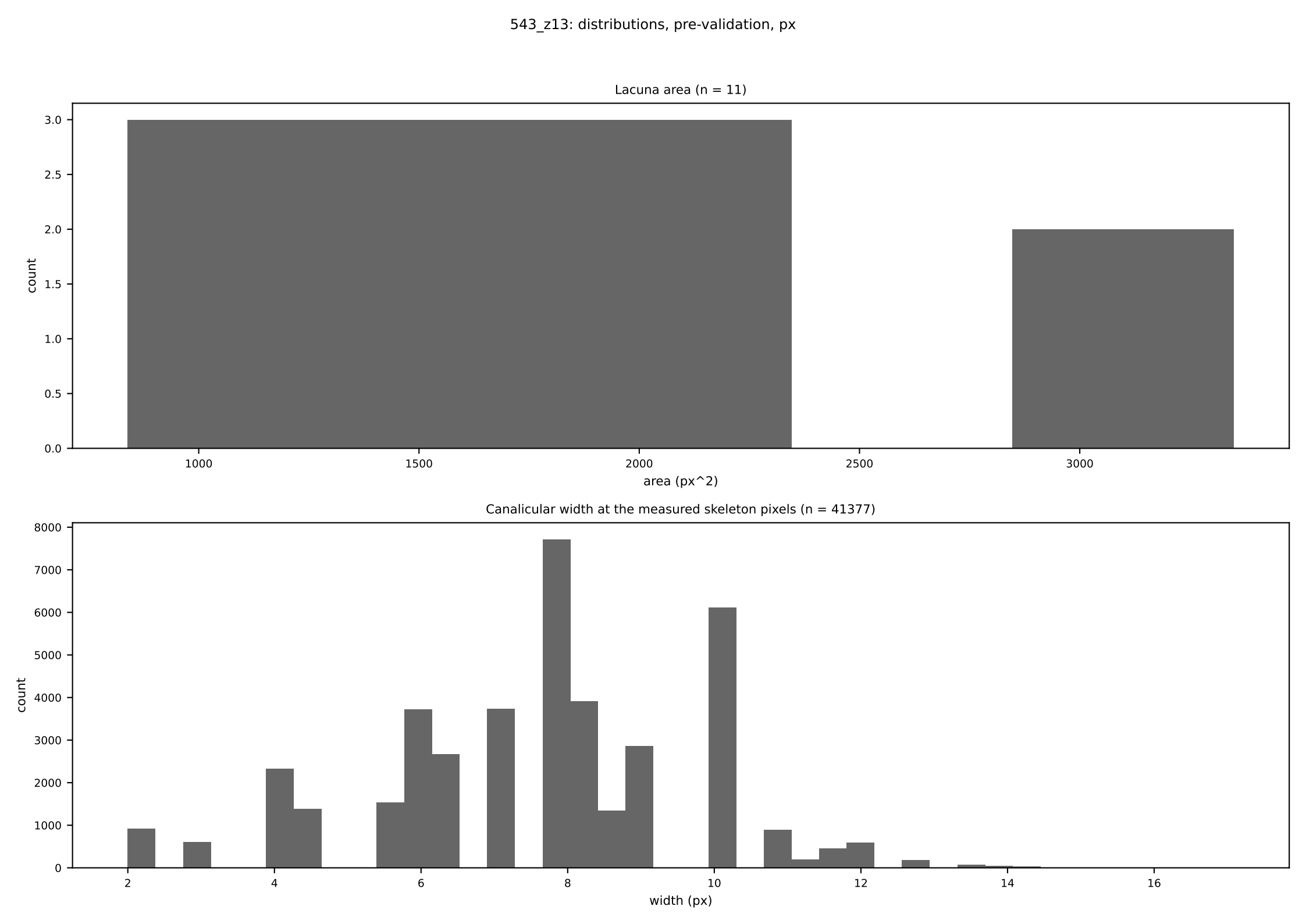
lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	274	826	353.889	1057.05	9	11	10
2	274	740	391.279	1062.28	6	13	14
3	260	523	376.534	993.749	11	13	12
4	147	460	276.108	972.151	4	8	13
5	286	784	389.321	1168.36	7	12	17
6	310	760	392.522	1055.01	5	13	15
7	248	502	379.463	1187.97	7	12	13
8	228	855	315.907	1017.42	8	10	15
9	156	498	263.066	859.252	4	8	12
10	188	333	270.581	788.541	5	7	11
11	330	966	474.233	1360.82	9	15	16


543_z13: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 41377)
count
width (px)